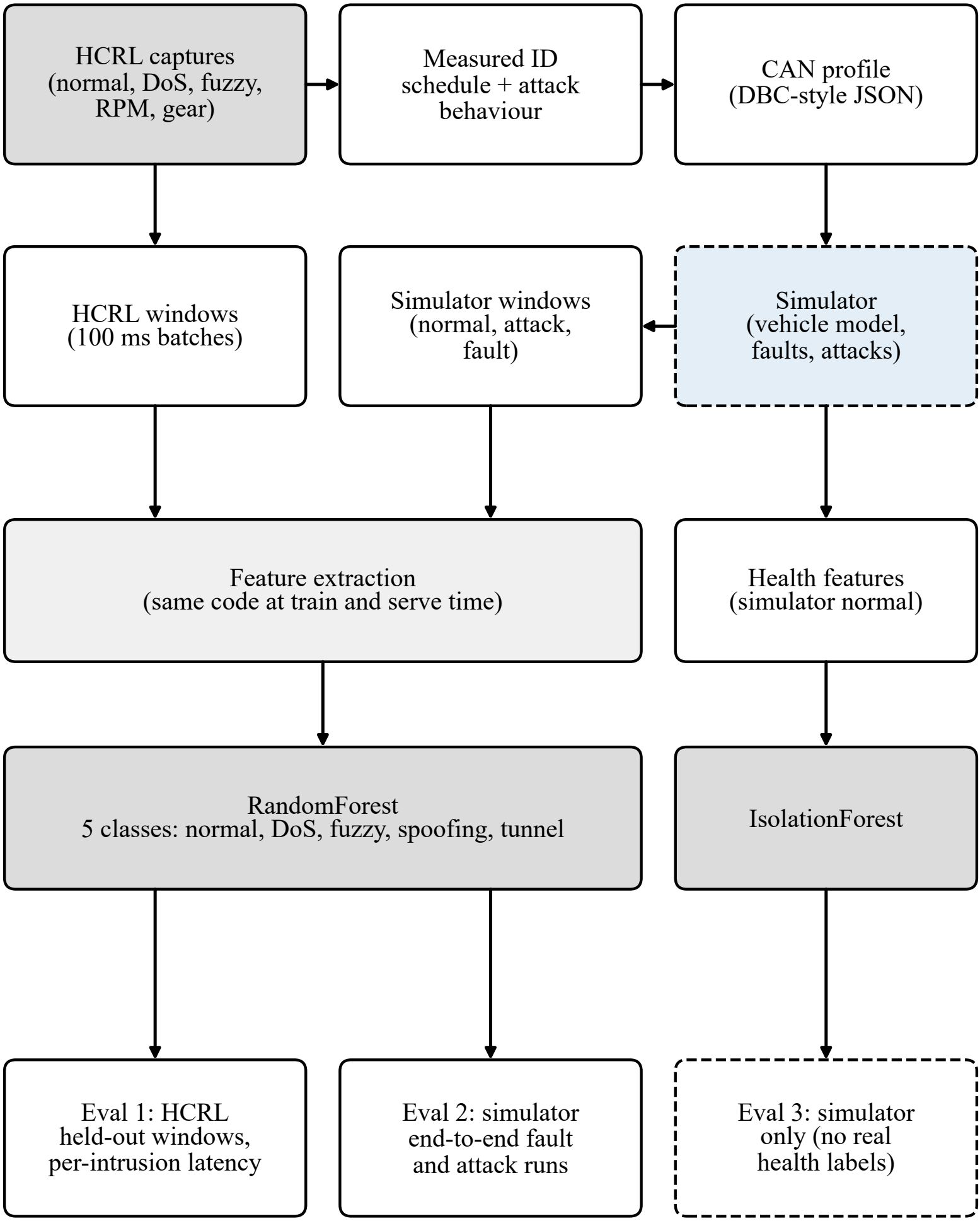
HCRL captures (normal, DoS, fuzzy, RPM, gear)
Measured ID schedule + attack behaviour
CAN profile (DBC-style JSON)
HCRL windows (100 ms batches)
Simulator windows (normal, attack, fault)
Simulator (vehicle model, faults, attacks)
Feature extraction (same code at train and serve time)
Health features (simulator normal)
RandomForest 5 classes: normal, DoS, fuzzy, spoofing, tunnel
IsolationForest
Eval 1: HCRL held-out windows, per-intrusion latency
Eval 2: simulator end-to-end fault and attack runs
Eval 3: simulator only (no real health labels)